數統導論作業四

111020003 林群佑

3.1.12

(a)

$$
\begin{aligned}
M_{X_2,\ldots,X_{k-1}}(t_2,\ldots,t_{k-1}) &= M_{X_1,\ldots,X_{k-1}}(0,t_2,\ldots,t_{k-1}) \\
&= (p_1e^0 + p_2e^{t_2} + \cdots + p_{k-1}e^{t_{k-1}} + p_k)^n \\
&= (p_1 + p_k + p_2e^{t_2} + \cdots + p_{k-1}e^{t_{k-1}})^n
\end{aligned}
$$

(b)

By mgf of (a) we can know that pmf of $X_2,\ldots,X_{k-1}$ is

$$
P(X_2 = x_2,\ldots,X_{k-1} = x_{k-1}) = \frac{n!}{x_2!\cdots x_{k-1}!(n-S)!}p_2^{x_2}\cdots p_{k-1}^{x_{k-1}}(p_1+p_k)^{n-s}
$$

where $S = \sum_{i=2}^{k-1} x_i \leq n$。

(c)

By definition

$$
\begin{aligned}
P(X_1 = x_1|X_2 = x_2,\ldots,X_{k-1} = x_{k-1}) &= \frac{P(X_1 = x_1, X_2 = x_2,\ldots,X_{k-1} = x_{k-1})}{P(X_2 = x_2,\ldots,X_{k-1} = x_{k-1})} \\
&= \frac{\frac{n!}{x_1!x_2!\cdots x_{k-1}!x_k!}p_1^{x_1}p_2^{x_2}\cdots p_{k-1}^{x_{k-1}}p_k^{x_k}}{\frac{n!}{x_2!\cdots x_{k-1}!(x_1+x_k)!}p_2^{x_2}\cdots p_{k-1}^{x_{k-1}}(p_1+p_k)^{(x_1+x_k)}} \\
&= \frac{(x_1+x_k)!}{x_1!x_k!}\frac{p_1^{x_1}p_k^{x_k}}{(p_1+p_k)^{(x_1+x_k)}} \\
&= \frac{(n-S)!}{x_1!(n-S-x_1)!}\left(\frac{p_1}{p_1+p_k}\right)^{x_1}\left(\frac{p_k}{p_1+p_k}\right)^{n-S-x_1}
\end{aligned}
$$

where $S = \sum_{i=2}^{k-1} x_i$

(d)

By (c) we can know that $X_1|X_2 = x_2,\ldots,X_{k-1} = x_{k-1} \sim \text{Binomial}(n - \sum_{i=2}^{k-1} x_i\,,\frac{p_1}{p_1+p_k})$
Therefore,

$$
E(X_1|X_2 = x_2,\ldots,X_{k-1}) = \left(n - \sum_{i=2}^{k-1} x_i\right)\left(\frac{p_1}{p_1+p_k}\right)
$$

3.1.17

First we compute mgf of Geometric(p) only consider fail times before success.

$$\begin{aligned}
\mathbb{E}[e^{tx}] &= \sum_{x=0}^{\infty} e^{tx}\,(1-p)^x\,p \\
&= p\sum_{x=0}^{\infty}\left[e^t\,(1-p)\right]^x \\
&= \frac{p}{1-(1-p)\,e^t}
\end{aligned}$$

Because it need converge that $t < -\ln(1-p)$
Then in fact that we can seen Negative Binomial(r, p) as r times independent Geometric(p).
Therefore,

$$\begin{aligned}
M_{NB(r,p)}(t) &= \prod_{i=1}^{r} M_{Geo(p)}(t) \\
&= \left[\frac{p}{1-(1-p)\,e^t}\right]^r \\
&= p^r\left[1-(1-p)\,e^t\right]^{-r}
\end{aligned}$$

It also need satisfy that $t < -\ln(1-p)$
Expection of Negative Binomial is:

$$\begin{aligned}
\mathbb{E}[X] &= \frac{d}{dt}M(0) \\
&= \frac{d}{dt}\,p^r\left[1-(1-p)\,e^t\right]^{-r} \\
&= p^r\cdot(-r)\cdot\left[1-(1-p)\,e^t\right]^{-(r+1)}\cdot[-(1-p)e^t] \\
&= p^r\cdot(-r)\cdot\left[1-(1-p)\,e^0\right]^{-(r+1)}\cdot[-(1-p)e^0] \\
&= p^r\cdot r\cdot p^{-(r+1)}\cdot(1-p) \\
&= r\frac{(1-p)}{p}
\end{aligned}$$

Variance of Negative Binomial is:

$$\begin{aligned}
Var(X) &= \frac{d^2}{dt^2}\,\ln\left[M(0)\right] \\
&= \frac{d^2}{dt^2}\,\left[r\ln(p) - r\ln(1-(1-p)e^t)\right] \\
&= \frac{d}{dt}\,\left[\frac{r(1-p)e^t}{1-(1-p)e^t}\right] \\
&= \frac{r(1-p)e^t[1-(1-p)e^t] - r(1-p)e^t[-(1-p)e^t]}{[1-(1-p)e^t]^2} \\
&= \frac{r(1-p)e^0[1-(1-p)e^0] + r(1-p)e^0[(1-p)e^0]}{[1-(1-p)e^0]^2} \\
&= r\frac{(1-p)}{p^2}
\end{aligned}$$

3.1.27

By definition we have:

$$\begin{aligned}
P(X \geq k+j \mid X \geq k) &= \frac{P(X \geq k+j \cap X \geq k)}{P(X \geq k)} \\
&= \frac{P(X \geq k+j)}{P(X \geq k)} \\
&= \frac{\sum_{i=k+j}^{\infty}(1-p)^i p}{\sum_{m=k}^{\infty}(1-p)^m p} \\
&= \frac{(1-p)^{k+j}}{(1-p)^k} \\
&= (1-p)^j \\
&= \sum_{i=j}^{\infty}(1-p)^i p \\
&= P(X \geq j)
\end{aligned}$$

3.2.17

Because X_1, X_2 are independent and $Y = X_1 + X_2$. Then we have $M_Y(t) = M_{X_1}(t)M_{X_2}(t)$.

$$\begin{aligned}
M_{X_2}(t) &= \frac{M_Y(t)}{M_{X_1}(t)} \\
&= \frac{exp[\mu(e^t-1)]}{exp[\mu_1(e^t-1)]} \\
&= exp[(\mu-\mu_1)(e^t-1)]
\end{aligned}$$

By mgf of X_2 and $\mu > \mu_1$. We can identify that $X_2 \sim Possion(\mu - \mu_1)$.

3.3.16

$$\begin{aligned}
P(Y < y) &= P(-2\ln(X) < y) \\
&= P(\ln(X) > -\frac{y}{2}) \\
&= P(X > e^{-y/2}) \\
&= \int_{e^{-y/2}}^{1} 1\,dx \\
&= 1 - e^{-y/2}
\end{aligned}$$

Then when $X = 0, Y = \infty$ and $X = 1, Y = 0$

Therefore, we have cdf of Y:

$$F_Y(y) = \begin{cases} 1 - e^{-y/2} \text{ if } y \in (0,\infty) \\ 0 \text{ elsewhere} \end{cases}$$

$$\begin{aligned}
\frac{d}{dy}F_Y(y) &= \frac{d}{dy}1 - e^{-y/2} \\
&= \frac{1}{2}e^{-y/2}
\end{aligned}$$

The pdf of Y:

$$f_Y(y) = \begin{cases} \frac{1}{2}e^{-y/2} \text{ if } y \in (0,\infty) \\ 0 \text{ elsewhere} \end{cases}$$

3.3.24

(a)

Because X_1, X_2 are independent.

$$\begin{aligned} M_Y(t) &= \mathbb{E}[e^{tY}] \\ &= \mathbb{E}[e^{t(2X_1+6X_2)}] \\ &= \mathbb{E}[e^{2tX_1}]\mathbb{E}[e^{6tX_2}] \\ &= M_{X_1}(2t)M_{X_2}(6t) \\ &= [1-3(2t)]^{-3} \cdot [1-1(6t)]^{-5} \\ &= (1-6t)^{-8} \end{aligned}$$

(b)

By mgf of Y,we can know that $Y \sim Gamma(\alpha = 8, \beta = 6)$.

3.3.25

(a)

$$\begin{aligned} P(X > x + y \mid X > x) &= \frac{P(X > x + y \cap X > x)}{P(X > x)} \\ &= \frac{P(X > x + y)}{P(X > x)} \\ &= \frac{1 - F_X(x + y)}{1 - F_X(x)} \\ &= \frac{e^{-\lambda(x+y)}}{e^{-\lambda x}} \\ &= e^{-\lambda y} \\ &= 1 - F_X(y) \\ &= P(X > y) \end{aligned}$$

(b)

Let $g(y) = 1 - F_Y(y) = P(Y > y)$.The memoryless property states:

$$P(Y > x + y \mid Y > x) = P(Y > y)$$

Using the definition of conditional probability:

$$\frac{P(Y > x + y \cap Y > x)}{P(Y > x)} = \frac{P(Y > x + y)}{P(Y > x)} = P(Y > y)$$

Substituting $g(y)$:

$$\frac{g(x+y)}{g(x)} = g(y) \implies g(x+y) = g(x)g(y)$$

This is Cauchy's functional equation. Since Y is a continuous random variable, $F_Y(y)$ is continuous, and thus $g(y)$ must be continuous. The only continuous solutions for $y > 0$ are of the form $g(y) = c^y$ for some constant c.
We are given $0 < F_Y(y) < 1$ for $y > 0$, which implies $0 < g(y) < 1$ for $y > 0$.This restricts the constant c to $0 < c < 1$.
We can define a positive constant $\lambda = -\ln(c)$ (which is positive since $0 < c < 1$). Then $c = e^{-\lambda}$.
So, $g(y) = c^y = (e^{-\lambda})^y = e^{-\lambda y}$.
Finally, we substitute back $g(y) = 1 - F_Y(y)$:

$$1 - F_Y(y) = e^{-\lambda y}$$

$$F_Y(y) = 1 - e^{-\lambda y}, \quad \text{for } y > 0$$

This is the cdf of an exponential distribution with parameter λ. We also verify $F_Y(0) = 1 - e^0 = 0$, which matches the given condition.

3.4.29

Because X_1 and X_2 are independent with normal distributions $N(6, 1)$ and $N(7, 1)$.
Therefore, we have:

$$X_1 - X_2 \sim N(6 - 7, 1 + 1) \sim N(-1, 2)$$

Then,

$$\begin{aligned} P(X_1 > X_2) &= P(X_1 - X_2 > 0) \\ &= 1 - \Phi(\frac{0 - (-1)}{\sqrt{2}}) \\ &= \Phi(-\frac{1}{\sqrt{2}}) \\ &\approx 0.23975 \end{aligned}$$

3.5.7

Because X and Y have a bivariate normal distribution, we have $Y|X = x$:

$$Y|X = x \sim N(\mu_2 - \rho(x - \mu_1), \sigma_2^2(1 - \rho^2)) \sim N(10 - \rho(x - 5), 25(1 - \rho^2))$$

Then,

$$\begin{aligned} P(4 < Y < 16 \mid X = 5) &= \Phi\left(\frac{16 - \rho(5 - 5) - 10}{\sqrt{25(1 - \rho^2)}}\right) - \Phi\left(\frac{4 - \rho(5 - 5) - 10}{\sqrt{25(1 - \rho^2)}}\right) \\ &= \Phi\left(\frac{6}{5\sqrt{1 - \rho^2}}\right) - \Phi\left(\frac{-6}{5\sqrt{1 - \rho^2}}\right) \\ &= 2\Phi\left(\frac{6}{5\sqrt{1 - \rho^2}}\right) - 1 \end{aligned}$$

And we know that

$$P(4 < Y < 16 \mid X = 5) = 0.954 \approx 2\Phi(2) - 1$$

Therefore

$$\frac{6}{5\sqrt{1 - \rho^2}} \approx 2 \Rightarrow \rho \approx 0.8$$

3.5.12

Because X and Y have a bivariate normal distribution with parameters $\mu_1 = \mu_2 = 0$, $\sigma_1^2 = \sigma_2^2 = 1$, and correlation coefficient ρ.

$$Z = aX + bY \sim N(a\mu_1 + b\mu_2, a^2\sigma_1^2 + b^2\sigma_2^2 + 2ab\rho\sigma_1\sigma_2) \sim N(0, a^2 + b^2 + 2ab\rho)$$

3.6.11

A random variable F follows an F-distribution with r_1 and r_2 degrees of freedom, denoted $F \sim F(r_1, r_2)$, if it can be written as the ratio of two independent chi-square (χ^2) random variables, each divided by its respective degrees of freedom:

$$F = \frac{U_1/r_1}{U_2/r_2}$$

Where: $U_1 \sim \chi^2_{r_1}$, $U_2 \sim \chi^2_{r_2}$, U_1 and U_2 are independent. Then

$$T^2 = \frac{W^2/1}{V/r}$$

1. The Numerator (U_1/r_1):
   - Let $U_1 = W^2$ and $r_1 = 1$.
   - We are given $W \sim N(0, 1)$.
   - By definition, the square of a standard normal random variable is a chi-square random variable with 1 degree of freedom.
   - Therefore, $U_1 = W^2 \sim \chi^2_1$.
   - This matches the requirement for the numerator, with $r_1 = 1$.
2. The Denominator (U_2/r_2):
   - Let $U_2 = V$ and $r_2 = r$.
   - We are given $V \sim \chi^2_r$.
   - This matches the requirement for the denominator, with $r_2 = r$.
3. Independence:
   - We need U_1 and U_2 to be independent.
   - $U_1 = W^2$ and $U_2 = V$.
   - We are given that W and V are independent. Since W^2 is a function of W, it is also independent of V.
   - Therefore, U_1 and U_2 are independent.

We have successfully shown that T^2 can be expressed as:

$$T^2 = \frac{W^2/1}{V/r} = \frac{U_1/r_1}{U_2/r_2}$$

Where: $U_1 = W^2 \sim \chi^2_1$ (so $r_1 = 1$) , $U_2 = V \sim \chi^2_r$ (so $r_2 = r$), U_1 and U_2 are independent. This is the exact definition of an F-distribution with $r_1 = 1$ and $r_2 = r$. Thus, $T^2 \sim F(1, r)$.